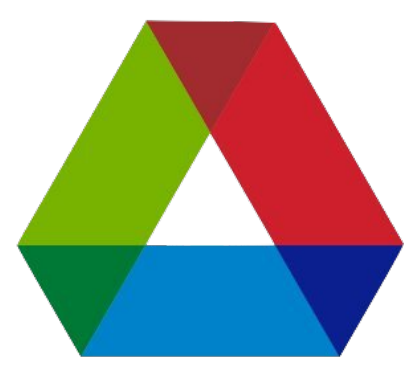




Kolmogorov-Arnold Neural Networks as Implicit Neural Representations for Volumetric Data

Charles O'Hanlon¹ Mengjiao Han¹ Joseph Insley¹ Janet Knowles¹ Victor Mateevitsi¹ Michael E. Papka¹
Silvio Rizzi¹ Qi Wu²

¹Argonne National Laboratory ²Nvidia

Motivation

INRs as lossy compression.

An implicit neural representation (INR) stores only the parameters θ of a continuous function $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^c$ that can be queried at arbitrary resolution, trading some fidelity for significant storage savings.

Volumetric case.

For volumetric data we learn $f_\theta : \mathbb{R}^3 \rightarrow \mathbb{R}$, mapping spatial coordinates to scalar values. Downstream algorithms (e.g. differentiable rendering) can query f_θ directly without decoding a dense grid.

Why KANs?

Kolmogorov-Arnold Networks substitute spline or spline-like edge activations for linear weight matrices, yielding localized basis functions that are directly interpretable.

Future work.

By visualising or manipulating these latent spline functions we aim to discover structural features or perform coarse edits without the costly step of full volume reconstruction.

KAN Architecture and Universal Approximation

Traditional MLP universal approximation theorem (Cybenko 1989):

$$f(\mathbf{x}) \approx \sum_{i=1}^{N_c} a_i \sigma(\mathbf{w}_i \cdot \mathbf{x} + b_i).$$

Kolmogorov-Arnold representation theorem (KAN formulation):

$$f(x_1, \dots, x_n) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^n \varphi_{q,p}(x_p) \right),$$

with learnable univariate functions $\varphi_{q,p}$ and Φ_q . Stacking yields a deep KAN:

$$\text{KAN}(\mathbf{x}) = (\Phi_L \circ \dots \circ \Phi_2 \circ \Phi_1)(\mathbf{x}).$$

Layer implementation.

- Edge activations: cubic B-splines with trainable control points.
- Fast KAN: replaces each B-spline with a radial basis function approximation, removing the spline evaluation bottleneck and cutting training time by roughly 1.5-2 x. All experiments use Fast KAN.
- Node outputs are sums of incoming edge activations.
- Sparse banded Jacobians keep back-prop memory efficient.

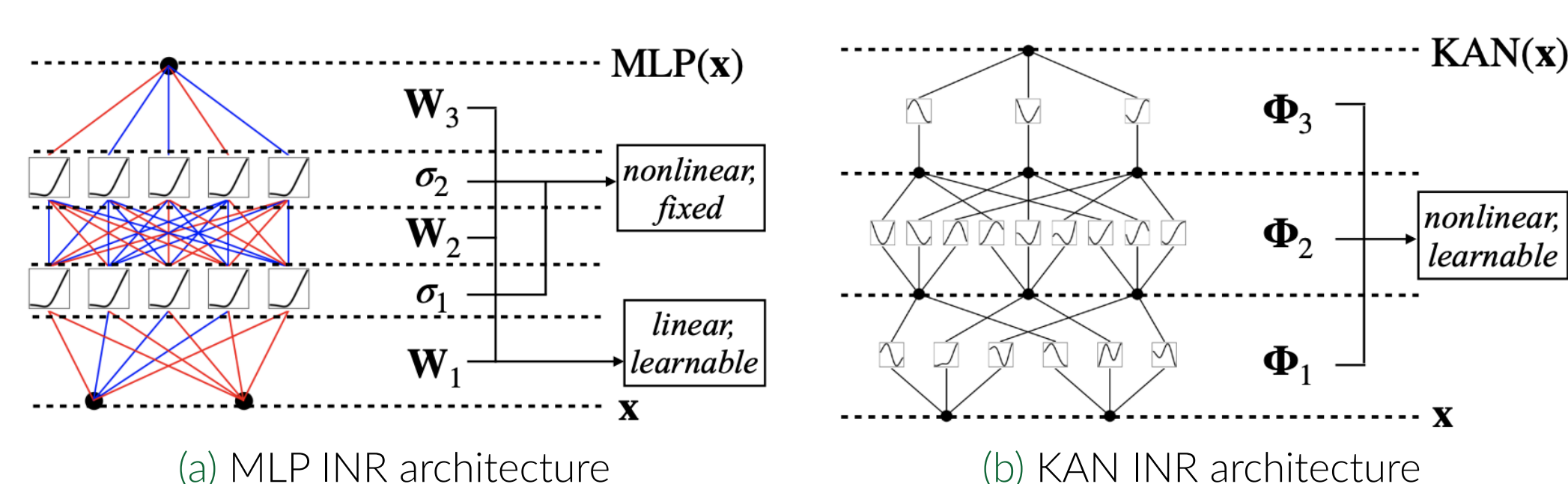


Figure 1. Implicit representation architectures: MLP (left) versus KAN (right).

Experimental Setup

- Datasets** (Klaccansky Open SciVis):
 - Nucleon - 41 x 41 x 41, uint8, spacing 1 x 1 x 1.
 - Neghip - 64 x 64 x 64, uint8, spacing 1 x 1 x 1.
 - MRT Angio - 416 x 512 x 112, uint16, spacing 0.412 mm isotropic.
- Model grid:** 4 widths {64, 32, 16, 8} x 3 hidden layers for both MLP and Fast KAN (8 configs total).
- Training:** batch size 32768, 16 epochs (≈ 50 k steps); Adam (beta1 0.9, beta2 0.999, lr 1e-3) with cosine decay; L2 voxel loss.
- Hardware:** 1 Sophia node, 32 CPU cores, 8 A100 GPUs (40 GB each).
- Metrics:** PSNR, SSIM, and MSE recorded for reconstructed dataset during evaluation step.

KAN vs. MLP Across Metrics and Volumes

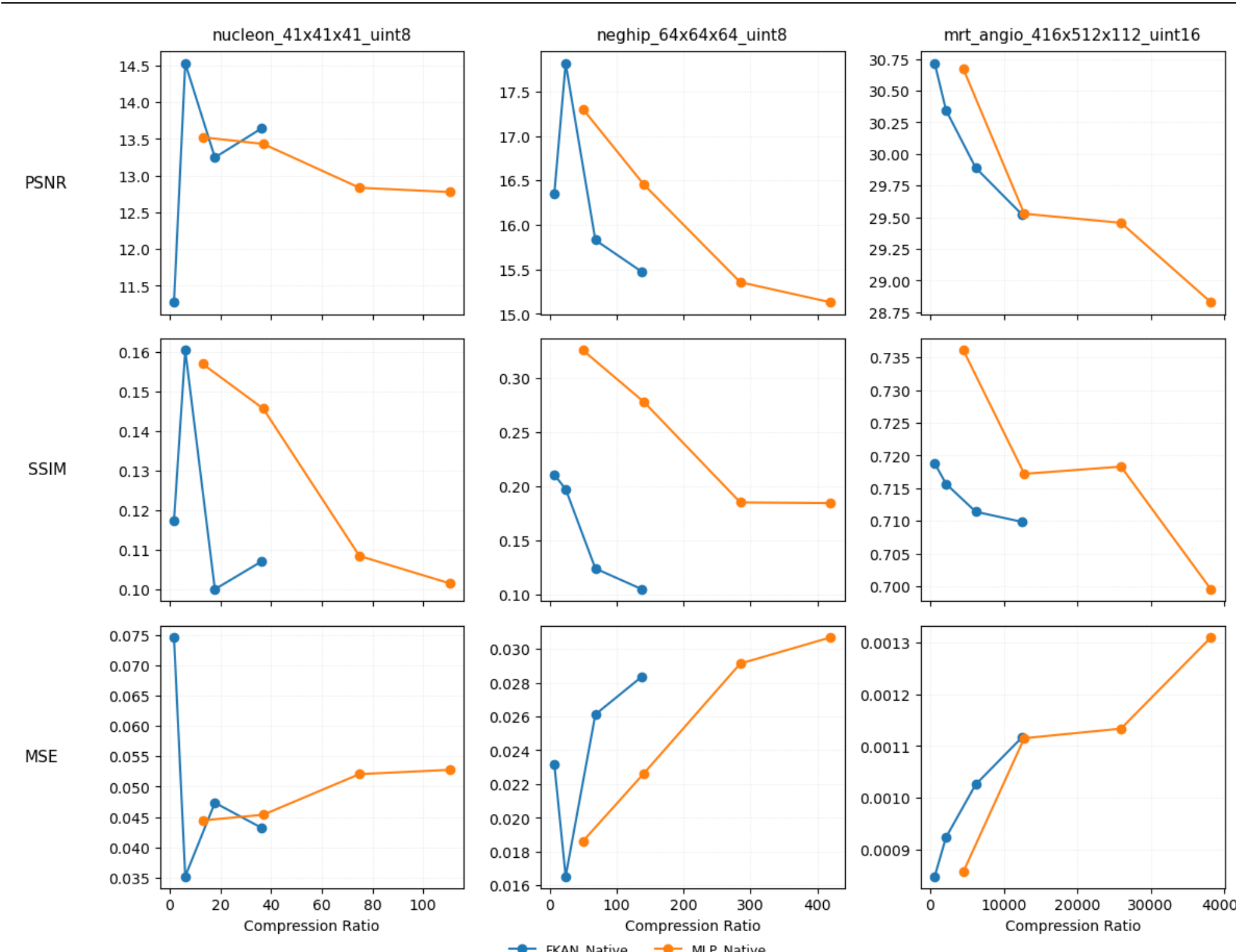


Figure 2. Compression sweeps: rows = PSNR, SSIM, MSE; columns = datasets. Curves: Fast KAN (blue) vs. MLP (orange).

Results and Discussion

Overall trends. Across all volumes Fast KAN tends to under-perform in the high-fidelity, low-compression regime (compression ratio < 30), reflecting its stronger expressivity per parameter. As compression increases, KANs narrow the gap and occasionally overtake MLPs on the smaller Nucleon and Neghip volumes, whereas MLP maintains a clear advantage on the large MRT Angio scan.

Takeaways. Fast MLP is preferable when reconstruction fidelity is critical or when volumes are large. For aggressive compression on small volumes a KAN of matching size can be competitive. These benchmarks establish a quantitative baseline for future INR research on scientific data.

Peak Signal to Noise Ratio (PSNR)

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right) \text{ dB}, \quad \text{MAX} = 2^B - 1.$$

Intuition: logarithmic fidelity measure; higher is better.

Structural Similarity Index (SSIM)

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}.$$

Intuition: perceptual similarity combining luminance, contrast, and structure; ranges in [0,1].

Mean Squared Error (MSE)

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (x_i - y_i)^2.$$

Intuition: average squared voxel error; lower is better.

KAN Interpretability and Future Work

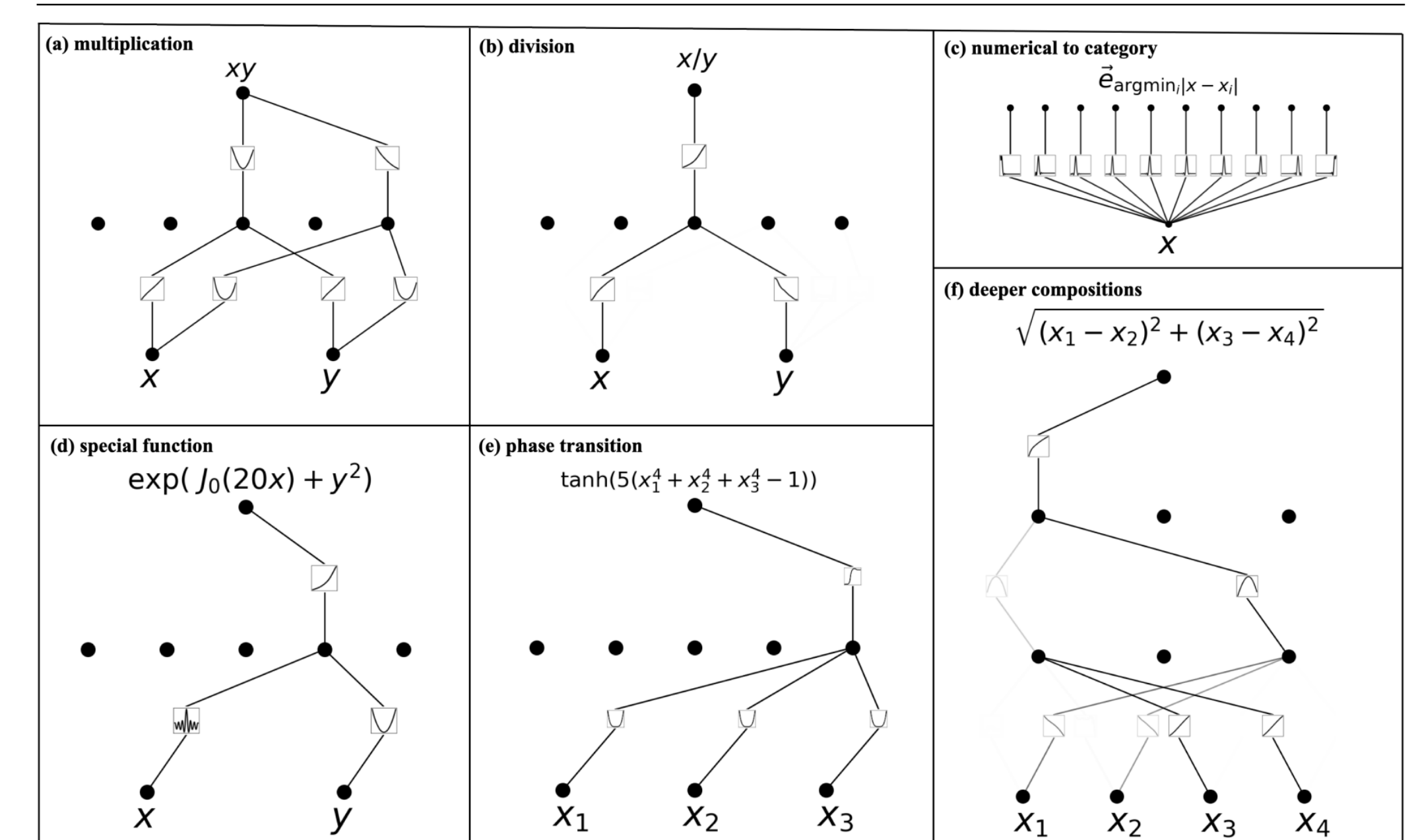


Figure 3. Symbolic concepts recovered by pruning and visualising KAN edges: (a) multiplication, (b) division, (c) nearest category, (d) special function, (e) phase transition, (f) deeper composition.

- Spline plotting:** inspect learned $\varphi_{q,p}$ functions to highlight salient spatial ranges.
- Edge pruning:** inactive splines zero out, yielding sparse analytic forms.
- Latent edits:** modifying a handful of spline control points enables coarse structure changes without full-volume decoding.

Acknowledgements

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